

Hard questions 1

1.

Function f is defined by $f(x) = -a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x) - 12$ has a y -intercept at $(0, -\frac{75}{7})$. The product of a and b is $\frac{320}{7}$. What is the value of a ?

2.

A certain town has an area of **4.36** square miles. What is the area, in square yards, of this town?
(1 mile = 1,760 yards)

A. 404

B. 7,674

C. 710,459

D. 13,505,536

3.

x	y
21	-8
23	8
25	-8

The table shows three values of x and their corresponding values of y , where $y = f(x) + 4$ and f is a quadratic function. What is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?

4.

For groups of **25** or more people, a museum charges **\$21** per person for the first **25** people and **\$14** for each additional person. Which function ***f*** gives the total charge, in dollars, for a tour group with ***n*** people, where ***n* ≥ 25**?

A. $f(n) = 14n + 175$

B. $f(n) = 14n + 525$

C. $f(n) = 35n - 350$

D. $f(n) = 14n + 21$

5.

$$\begin{aligned}\frac{3}{2}y - \frac{1}{4}x &= \frac{2}{3} - \frac{3}{2}y \\ \frac{1}{2}x + \frac{3}{2} &= py + \frac{9}{2}\end{aligned}$$

In the given system of equations, ***p*** is a constant. If the system has no solution, what is the value of ***p***?

6.

In the *xy*-plane, a parabola has vertex **(9, −14)** and intersects the *x*-axis at two points. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where ***a***, ***b***, and ***c*** are constants, which of the following could be the value of ***a + b + c***?

A. **−23**

B. **−19**

C. **−14**

D. **−12**